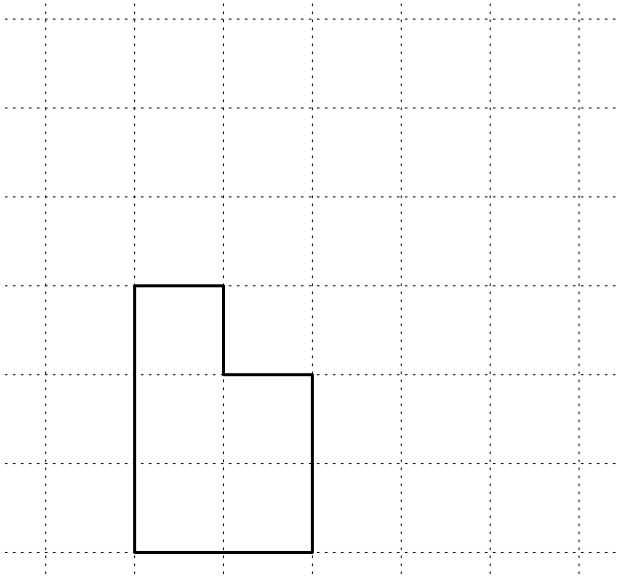
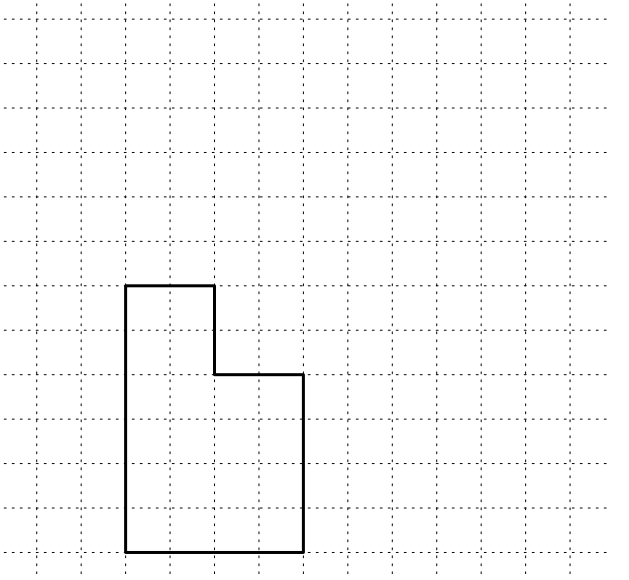
(1) original



(2) scale k=2